

Matrix: A $p \times m$

$$A = \begin{bmatrix} \underline{a}_{11} & \underline{a}_{12} & \dots & \underline{a}_{1m} \\ \underline{a}_{21} & \underline{a}_{22} & \dots & \underline{a}_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \underline{a}_{p1} & \underline{a}_{p2} & \dots & \underline{a}_{pm} \end{bmatrix}$$

$\underline{\alpha}_1^T, \underline{\alpha}_2^T, \dots, \underline{\alpha}_p^T$ $\underline{\alpha}_i \in \mathbb{R}^m$
ROWS of A

$\underline{a}_1, \underline{a}_2, \dots, \underline{a}_m$ $\underline{a}_j \in \mathbb{R}^p$
columns of A

$$A = [\underline{a}_1, \underline{a}_2, \dots, \underline{a}_m] \text{ Column partition}$$

$$= \begin{bmatrix} \underline{\alpha}_1^T \\ \underline{\alpha}_2^T \\ \vdots \\ \underline{\alpha}_p^T \end{bmatrix} \text{ row partition.}$$

Matrix-vector multi:

A $p \times m$, $x \in \mathbb{R}^m$, product

$$y = Ax$$

Defined only if $n=m$, and then

$y \in \mathbb{R}^p$ given by

$$y = \sum_{j=1}^m x_j \underline{a}_j$$

(linear comb. of cols of A)

Equivalently

$$y = \begin{bmatrix} \underline{\alpha}_1 \cdot x \\ \underline{\alpha}_2 \cdot x \\ \vdots \\ \underline{\alpha}_p \cdot x \end{bmatrix} \text{ array of dot products with rows of } A$$

Range: A $p \times m$

$$R(A) = \text{span}\{\underline{a}_j\}_{j=1}^m \subset \mathbb{R}^p$$

Rank: $\rho(A) = \dim R(A)$

$$= \dim \text{span}\{\underline{a}_j\}_{j=1}^m$$

= # lin indep cols A

Nullspace

$$N(A) = \{x \in \mathbb{R}^m, Ax = \underline{0}\} (x \neq \underline{0})$$

$$N(A) \subset \mathbb{R}^m$$

Nullity:

$$\nu(A) = \dim N(A)$$

= max # lin indep vectors
in nullspace

Rank-Nullity Thm:

$$\rho(A) + \nu(A) = m \text{ (# cols of } A)$$

Matrix-matrix mult:

Given A $m \times n$, B $p \times q$ product
must match (inner)

$$C = AB \quad (m \times n)(p \times q)$$

size of (outer)

defined only if $n=p$, and then

$$C = A[b_1 \dots b_q]$$

$$= [(Ab_1) (Ab_2) \dots (Ab_q)]$$

has size $m \times q$

Equivalently $C_{ij} = \sum_{l=1}^n a_{il} b_{lj} \quad i=1, \dots, m \quad j=1, \dots, q$

(dot product of i^{th} row of A
with j^{th} col of B)

"Square" Matrix: Matrix has same
rows as cols

"Identity matrix": I $n \times n$
for any $x \in \mathbb{R}^n$, $Ix = x$
for any A $n \times n$ $IA = AI = A$

Matrix inverse: for A $n \times n$

Matrix M is an inverse if

$$MA = AM = I$$

If such an M exists we denote it by A^{-1}

Computationally $A^{-1} = \frac{\text{Adj}(A)}{\text{Det}(A)}$

Det scalar, Adj $n \times n$ matrix of sub-det
can be recursively reduced to 2×2 case

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\text{Det}(A) = a_{11}a_{22} - a_{21}a_{12}$$

$$\text{Adj}(A) = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Note: A^{-1} undefined if $\text{Det}(A) = 0$
Such a matrix is singular

FACT: $\text{Det}(A) = 0 \iff$

- A^{-1} DNE (singular)
- cols A Not linearly indep
- $\rho(A) < n$
- $\dim R(A) < n$
- $\rho(A) > 0$
- $N(A)$ nontrivial

Eigenvalues/Vectors:

A pair $\lambda \in \mathbb{C}$, $\underline{v} \in \mathbb{C}^n$, $\underline{v} \neq \underline{0}$ such that

$$A\underline{v} = \lambda \underline{v}$$

Equivalently: $(\lambda \mathbf{I} - A)\underline{v} = \underline{0}$

hence λ is such that $(\lambda \mathbf{I} - A)$ is singular
and $\underline{v} \in N(\lambda \mathbf{I} - A)$

$(\lambda \mathbf{I} - A)$ singular $\iff \text{Det}(\lambda \mathbf{I} - A) = 0$

$\text{Det}(\lambda \mathbf{I} - A) = p_A(\lambda)$ "Characteristic polynomial" of A

$$= \lambda^n + \alpha_{n-1}\lambda^{n-1} + \dots + \alpha_1\lambda + \alpha_0$$

Eigenvalues are roots of this poly
 $p_A(\lambda) = 0$ (char. eq'n)

$\implies n$ values satisfy $\lambda_1, \dots, \lambda_n$

for each eigenvalue λ_k ,
corresponding eigenvector $\underline{v}_k$ satisfies

$$(\lambda_k \mathbf{I} - A)\underline{v}_k = \underline{0} \quad k=1, \dots, n$$

Square, nonsingular matrices

Given A $n \times n$, $\text{Det}(A) \neq 0$

Then

$$\text{Span} \{ \underline{a}_k \}_{k=1}^n = \mathbb{R}^n$$

$\{ \underline{a}_k \}$ linearly indep

$\implies$ Cols of A provide a basis for $\mathbb{R}^n$

$$\implies R(A) = \mathbb{R}^n$$

$\implies$ any $\underline{y} \in \mathbb{R}^n$ admits expansion

$$\underline{y} = \sum_{k=1}^n x_k \underline{a}_k$$

x_k "coordinates" of $\underline{y}$ in basis $\{ \underline{a}_k \}$

But then $\underline{y} = A\underline{x}$

$$\implies \underline{x} = A^{-1}\underline{y}$$

$\implies$ Any nonsingular matrix defines a basis for $\mathbb{R}^n$.

Note: if $\lambda = 0$ is an eigenvalue of A then

$$\det(\lambda I - A) = \det(A) = 0$$

hence A is singular and its cols do not define a basis for $\mathbb{R}^n$.

Caley-Hamilton Thm

Every matrix satisfies its own char eq'n: $P_A(A) = O_{n \times n}$

$$\text{i.e. } A^n + \alpha_{n-1}A^{n-1} + \dots + \alpha_1 A + \alpha_0 I = O_{n \times n}$$

where $A^2 = A \cdot A$ etc.

and $\alpha_0, \alpha_1, \dots, \alpha_n$ are coeffs in char poly that defines the eigenvalues

Equivalently

$$A^n = -[\alpha_0 I + \alpha_1 A + \dots + \alpha_{n-1} A^{n-1}]$$

i.e. A^n is a linear comb of $A^0 = I$ up to A^{n-1}

Hence

$$A^{n+1} = -[\alpha_0 A + \alpha_1 A^2 + \dots + \alpha_{n-1} A^n]$$

can also be written as a linear comb of $A^0, \dots, A^{n-1}$ (subst from above)

By induction we have:

for any $K \geq n$

A^K can be written as a linear comb of A^j $0 \leq j \leq n-1$

= Return to linear system:

$$\dot{\underline{x}}(t) = A(t)\underline{x}(t) + B(t)\underline{u}(t)$$

$$\underline{x}(t_0) = \underline{x}_0$$

Problem: given any $\underline{x}_0 \in \mathbb{R}^n$, $\underline{u}(t) \in \mathbb{R}^m$ $t \geq t_0$

find $\underline{x}(t)$, $t \geq t_0$

Start w/ homog. eq'n

$$\dot{\underline{x}}(t) = A(t)\underline{x}(t), \quad \underline{x}(t_0) = \underline{x}_0$$

Suppose we know, or can easily find, n sol'n's $\underline{\psi}_k(t)$ from an arbitrary set of ICs. $\{\underline{\psi}_k(t_0)\}_{k=1}^n$ which are linearly indep.

i.e.

$$\dot{\underline{\psi}}_k(t) = A(t) \underline{\psi}_k(t) \quad k=1, \dots, n$$

$$\underline{\psi}_k(t_0) = \underline{\psi}_{k,0}$$

w/ $\underline{\psi}_{k,0}$ arbitrary, but lin. indep.

Any such set $\{\underline{\psi}_k(t)\}_{k=1}^n$ are known as fundamental sol'n's

and

$$\underline{\Psi}(t) = [\underline{\psi}_1(t) \quad \underline{\psi}_2(t) \quad \dots \quad \underline{\psi}_n(t)]$$

is known as a fundamental matrix

Note:

$$\begin{aligned} \dot{\underline{\Psi}}(t) &= [\dot{\underline{\psi}}_1(t) \quad \dots \quad \dot{\underline{\psi}}_n(t)] \\ &= [A \underline{\psi}_1(t) \quad \dots \quad A \underline{\psi}_n(t)] \\ &= A \underline{\Psi}(t) \end{aligned}$$

i.e. $\dot{\underline{\Psi}}(t) = A(t) \underline{\Psi}(t)$

w/ $\underline{\Psi}(t_0) = [\underline{\psi}_1(t_0) \quad \dots \quad \underline{\psi}_n(t_0)]$
nonsingular

A matrix version of the homog. state eq'n, with nonsingular B/c.

Claim: for any $\underline{x}_0 = \underline{x}(t_0) \in \mathbb{R}^n$ and any set of fund'l sol'n's $\{\underline{\psi}_k(t)\}$, the resulting $\underline{x}(t)$ can be written as a linear comb of $\{\underline{\psi}_k(t)\}$

i.e. $\underline{x}(t) = \sum_{k=1}^n \alpha_k \underline{\psi}_k(t) \quad \forall t \geq t_0$

PF: By def'n, $\{\underline{\psi}_k(t_0)\}$ form a basis for $\mathbb{R}^n$.

$\Rightarrow$ arbitrary $\underline{x}_0$ admits expansion

$$\underline{x}_0 = \sum_{k=1}^n \alpha_k \underline{\psi}_k(t_0)$$

so true for $t = t_0$. what about $t > t_0$

See if the assumption is consistent

$$\dot{\underline{x}}(t) = \sum_{k=1}^n \alpha_k \dot{\Psi}_k(t)$$

$$= \sum_{k=1}^n \alpha_k (A(t) \Psi_k(t))$$

$$= A(t) \sum_{k=1}^n \alpha_k \Psi_k(t)$$

$$= A(t) \underline{x}(t) \quad \checkmark$$

So $\underline{x}(t)$ satisfies the DE and B/c
By uniqueness it is the sol'n.

Note: Can write $\underline{x}(t)$ as

$$\underline{x}(t) = \underline{\Psi}(t) \underline{\alpha}$$

where $\underline{\Psi}(t)$ is fund matrix

$$\text{and } \underline{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

where $\underline{\alpha}$ also satisfies

$$\underline{x}_0 = \underline{\Psi}(t_0) \underline{\alpha}$$

$$\Rightarrow \underline{\alpha} = \underline{\Psi}(t_0)^{-1} \underline{x}_0$$

valid since $\underline{\Psi}(t_0)$ nonsingular

$$\Rightarrow \underline{\alpha} = \underline{\Psi}^{-1}(t) \underline{x}(t), \forall t \geq t_0$$

which would imply that $\underline{\alpha}$
is a "conserved" quantity
of the homog. dynamics -

Lemma 1: Any fund'l matrix
satisfies:

$$\frac{d}{dt} \text{Det } \underline{\Psi}(t) = \text{trace} \{A(t)\} \text{Det } \underline{\Psi}(t)$$

$$\text{trace} \{A(t)\} = \sum_{k=1}^n a_{kk}(t)$$

(sum of diagonal
elements)

Lemma 2: if $\int_{t_0}^t \text{trace}\{A(\tau)\} d\tau > -\infty$
for any finite $t \geq t_0$, then

$\Psi(t)$ is nonsingular for
any finite $t \geq t_0$.

Pf: Let $d(t) = \text{Det } \Psi(t)$
and $a(t) = \text{trace}\{A(t)\}$

Then Lemma 1 states

$$\dot{d}(t) = a(t)d(t)$$

scalar linear DE, with explicit
sol'n

$$d(t) = d(t_0) e^{\int_{t_0}^t a(\tau) d\tau}$$

$d(t) = 0 \Rightarrow$ either $d(t_0) = 0$

or $e^{\int_{t_0}^t a(\tau) d\tau} = 0$

But

$$d(t_0) = \text{Det } \Psi(t_0)$$

$\neq 0$ by assumption.

2nd requires

$$\int_{t_0}^t a(\tau) d\tau = -\infty$$

assuming this is not true, then

$d(t) \neq 0$ for any finite $t \geq t_0$

$\Rightarrow$ all "realistic" $A(t)$ will
satisfy req'd cond'n:

Hence

- $\text{Det } \Psi(t) \neq 0 \quad \forall t \geq t_0$

- $\Psi^{-1}(t)$ exists $\forall t \geq t_0$

- $\{\Psi_k(t)\}$ lin indep
 $\forall t \geq t_0$

Note:

$$\underline{x}(t) = \underline{\Psi}(t) \underline{\alpha} \quad \forall t \geq t_0$$

where $\underline{\alpha} = \underline{\Psi}(t_0)^{-1} \underline{x}_0$

Thus

$$\begin{aligned} \underline{x}(t) &= \underline{\Psi}(t) \underline{\Psi}^{-1}(t_0) \underline{x}_0 \quad \forall t \geq t_0 \\ &= \underline{\Phi}(t, t_0) \underline{x}_0 \end{aligned}$$

$\underline{\Phi}$ is the "state transition matrix" for the system; unique to system.

== For non-homog response ($\underline{u}(t) \neq \underline{0}$)
Then look at

$$\frac{d}{dt} [\underline{\Psi}^{-1}(t) \underline{x}(t)]$$